Time as Derived Structural Order in Darke Theory

Emergent Temporality from Recursive Differentiation, Closure, and Admissibility

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Abstract

Darke Theory does not permit temporal order to be assumed as a primitive at the foundational level. Stable structure, geometry, causality, and clock behaviour are treated as emergent consequences of recursive differentiation achieving closure under admissibility constraints. This paper develops a mathematically strengthened reconstruction of temporality within that framework. Time is defined not as a background parameter in which states evolve, but as the derived order induced by admissible, closure-bearing structural transition. Closure is formalised as a graded functional with constrained subcomponents. Admissibility is formalised as an indicator law governed by support, burden, persistence horizon, closure threshold, and reserve margin. Temporal precedence is defined as the reachability relation generated by admissible refinement and is given explicit order-theoretic structure. The arrow of time is derived from irreversibility under directional support asymmetry together with monotone growth of structural memory. Structural memory is decomposed into closure commitment, persistent residue, and branch-loss terms. Duration is defined as intrinsic path measure over admissible transition chains using an explicit transition-weight ansatz and an additive concatenation law. Clocks are interpreted as coarse-grained recurrent counters of approximately uniform local admissible reconfiguration, thereby avoiding conflict with the exclusion of nontrivial direction-bearing cycles. Time dilation is reconstructed as closure-density dependence of local transition cost. Structural age is then defined in path-aware form. The result is a disciplined account in which temporality is earned from persistence, closure, and memory rather than presupposed.

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1 Introduction

A foundational theory cannot legitimately presuppose as primitive any structure it claims to explain. If temporal order lies inside explanatory scope, then primitive time is inadmissible at the foundational layer. The same applies to primitive duration, primitive causal ordering, and primitive geometric background. Effective theories may deploy such notions successfully once stable structure already exists. A foundational theory may not.

Darke Theory begins from recursive differentiation rather than from spacetime. Persistent structure is not assumed. It is earned through recursive refinement achieving closure under admissibility constraints. The consequence is immediate: time cannot be treated as a background container in which change occurs. Temporal order must itself arise downstream of structural commitment.

The central claim of this paper is therefore:

Time is the measurable order of admissible structural commitment.

More precisely, the paper proceeds by:

1. formalising recursive refinement without primitive time,
2. formalising closure as a graded structural functional,
3. formalising admissibility as supportable persistence under reserve conditions,
4. deriving temporal precedence from admissible refinement reachability,

5. deriving temporal direction from irreversibility under directional asymmetry and structural memory growth,
6. deriving duration as intrinsic path measure,
7. deriving clocks as coarse-grained recurrent local counters,
8. reconstructing time dilation as closure-density dependence of local transition cost.

This is not an attempt to deny the success of relativity or any effective local dynamical description. It is an attempt to relocate their temporal notions onto a deeper ontological base.

2 Notation summary

For ease of reference, the principal symbols used throughout the paper are collected here.

Symbol	**Meaning**
$\mathcal{X}$	State space of refinement configurations.
$\preceq$	Refinement or structural-precedence partial order on $\mathcal{X}$.
D	Recursive substantiation operator on $\mathcal{X}$.
$\mathcal{L}_X$	Refinement limit associated with the recursive chain generated from X.
I_{self}	Internal self-integration subfunctional.
I_{rec}	Recursive invariance subfunctional.
I_{supp}	Support-stability subfunctional.
I_{pert}	Perturbation-resistance subfunctional.
$\mathsf{Cl}(X)$	Graded closure functional of state X.
$\mathsf{Cl}_{\text{crit}}$	Critical closure threshold defining the closed sector.
$\mathcal{C}_{\text{crit}}$	Closed sector $\{X \in \mathcal{X} : \mathsf{Cl}(X) \geq \mathsf{Cl}_{\text{crit}}\}$.
$\Delta\mathsf{Cl}(X, Y)$	Closure increment $\mathsf{Cl}(Y) - \mathsf{Cl}(X)$.
$\mathsf{S}(X \to Y; \ell)$	Available support for transition $X \to Y$ across persistence horizon ℓ.
$\mathsf{B}(X \to Y; \ell)$	Burden required to realise and maintain transition $X \to Y$ across horizon ℓ.
ℓ	Persistence horizon, span, or regime-specific extent relevant to a transition.
$\Pi(X \to Y)$	Reserve margin $\mathsf{S}(X \to Y; \ell) - \mathsf{B}(X \to Y; \ell)$.
$\pi_{\min}$	Minimum reserve threshold required for admissibility.
$\mathsf{Cl}_{\min}$	Minimum closure threshold required for admissibility.
$\mathcal{A}(X \to Y)$	Admissibility indicator for transition $X \to Y$.
$X \rightsquigarrow_A Y$	Admissible transition from X to Y.
$X \preccurlyeq_T Y$	Weak temporal precedence relation.
$X \prec_T Y$	Strict temporal precedence relation.
$\sim_A$	Admissible equivalence relation induced by mutual weak temporal precedence.
$\mathsf{Res}(X)$	Persistent structural residue encoded by state X.
$\mathsf{Br}(X)$	Irreversible branch-loss or excluded-possibility measure of state X.

Symbol	Meaning
$\mathsf{M}(X)$	Structural memory functional of state X.
$\Delta\mathsf{M}(X,Y)$	Memory increment $\mathsf{M}(Y)-\mathsf{M}(X)$.
$\Sigma(X,Y;\ell)$	Support surplus for transition $X \to Y$, identified with $\Pi(X \to Y)$.
$\Omega(X,Y;\ell)$	Directional asymmetry functional.
$\Xi(X,Y)$	Additional transition difficulty not already captured by closure or memory increments.
$w(X,Y)$	Local transition weight for a nontrivial admissible step.
$\tau(\gamma)$	Intrinsic duration of admissible path γ.
ω	Continuous transition-density functional in the continuous path analogue.
$X \sim_K Y$	Clock observational equivalence at the scale of clock readout.
$\delta\tau_K$	Intrinsic duration cost per coarse-grained clock cycle in subsystem K.
ρ_C	Local closure-density or closure-commitment environment.
$F(\rho_C)$	Closure-density rate-law factor governing cycle slowdown.
$\nu_K(\rho_C)$	Local clock-cycle frequency in subsystem K.
$\mathcal{P}(X_0,X)$	Family of admissible paths from reference state X_0 to X.
$\mathfrak{A}_{\min}(X)$	Minimal structural age of X.
$\mathfrak{A}_{\mathrm{can}}(X)$	Canonical structural age of X relative to a specified canonical path.
$\mathfrak{A}_{\exp}(X)$	Expected structural age of X under a measure on admissible paths.

3 Foundational discipline

The present construction stands downstream of the following Darke-Theoretic commitments.

Principle 1 (No primitive time)**.** Temporal order may not be assumed at the foundational layer if the theory claims to explain the conditions under which temporal order becomes meaningful.

Principle 2 (Closure threshold)**.** Recursive differentiation alone does not produce persistent structure. Closure is the threshold at which recursive refinement becomes self-integrating and representationally stable.

Principle 3 (Admissibility filter)**.** Not every formally definable branch is structurally realisable. A branch is physically meaningful only if it can be supported across its required persistence horizon without exhausting available support or collapsing below the required closure threshold.

These commitments force an inversion of the usual question. Instead of asking how states evolve in time, one must ask what structural relation takes over the work usually assigned to primitive time.

4 Recursive refinement systems

We begin with the substrate from which temporality will be derived.

Definition 1 (Recursive refinement system)**.** A recursive refinement system is a triple

$$(\mathcal{X}, \preceq, \mathsf{D}),$$

where:

i. $\mathcal{X}$ is a class of refinement states,

ii. $\preceq$ is a partial order representing refinement inclusion or structural precedence,

iii. $\mathsf{D} : \mathcal{X} \to \mathcal{X}$ is a recursive substantiation operator.

We assume throughout that $(\mathcal{X}, \preceq)$ is directed-complete on the sectors relevant to the present constructions and that D is monotone:

$$X \preceq Y \implies \mathsf{D}(X) \preceq \mathsf{D}(Y).$$

Where needed we further assume Scott continuity:

$$\mathsf{D}\left(\sup_i X_i\right) = \sup_i \mathsf{D}(X_i)$$

for directed families $\{X_i\}$.

Definition 2 (Recursive chain)**.** Given $X \in \mathcal{X}$, its recursive chain is

$$X,\ \mathsf{D}(X),\ \mathsf{D}^2(X),\ \ldots$$

or more generally any chain

$$X_0 \prec X_1 \prec X_2 \prec \cdots$$

compatible with $\preceq$ and generated by refinement.

Definition 3 (Refinement limit)**.** If the supremum exists, define

$$\mathcal{L}_X := \sup_{n \in \mathbb{N}} \mathsf{D}^n(X).$$

Lemma 1. *If D is Scott continuous, then $\mathcal{L}_X$ is a fixed point of D:*

$$\mathsf{D}(\mathcal{L}_X) = \mathcal{L}_X.$$

Proof. By Scott continuity,

$$\mathsf{D}(\mathcal{L}_X) = \mathsf{D}\left(\sup_n \mathsf{D}^n(X)\right) = \sup_n \mathsf{D}^{n+1}(X) = \mathcal{L}_X.$$

This does not yet produce time. It produces structurally grounded succession without presupposed temporality.

5 Closure as a graded structural functional

The first major requirement is that closure be a genuine mathematical object rather than a verbal label.

Definition 4 (Closure subfunctionals)**.** Let

$$I_{\text{self}},\ I_{\text{rec}},\ I_{\text{supp}},\ I_{\text{pert}} : \mathcal{X} \to [0, 1]$$

be the subfunctionals representing:

i. internal self-integration,

ii. recursive invariance under D,

iii. support stability across admissible continuation,

iv. perturbation resistance.

Remark 1. Normalising the subfunctionals to $[0, 1]$ prevents uncontrolled scaling and makes closure comparisons well-posed at the framework level.

Definition 5 (Closure functional)**.** A closure functional is a map

$$\mathsf{Cl} : \mathcal{X} \to [0, \infty)$$

defined by

$$\mathsf{Cl}(X) = \alpha_c I_{\text{self}}(X) + \beta_c I_{\text{rec}}(X) + \gamma_c I_{\text{supp}}(X) + \delta_c I_{\text{pert}}(X),$$

where $\alpha_c, \beta_c, \gamma_c, \delta_c > 0$.

Lemma 2. *For every $X \in \mathcal{X}$,*

$$0 \leq \mathsf{Cl}(X) \leq \alpha_c + \beta_c + \gamma_c + \delta_c.$$

Proof. Each subfunctional lies in $[0, 1]$ and each coefficient is positive.

Definition 6 (Closed sector)**.** For a critical closure threshold $\mathsf{Cl}_{\text{crit}} > 0$, define

$$\mathcal{C}_{\text{crit}} := \{X \in \mathcal{X} : \mathsf{Cl}(X) \geq \mathsf{Cl}_{\text{crit}}\}.$$

Definition 7 (Closure increment)**.** For a transition $X \to Y$, define

$$\Delta\mathsf{Cl}(X, Y) := \mathsf{Cl}(Y) - \mathsf{Cl}(X).$$

Principle 4 (Closure threshold for temporal meaning)**.** A transition carries physical temporal significance only if it participates in the formation, preservation, deformation, or loss of states in $\mathcal{C}_{\text{crit}}$.

6 Support, burden, horizon, and reserve

Closure is necessary but not sufficient. A candidate branch must also be supportable across its required persistence horizon.

Definition 8 (Support)**.** Support is the organisational resource available to maintain a candidate transition or regime across its relevant persistence horizon:

$$\mathsf{S}(X \to Y; \ell) \in [0, \infty).$$

Definition 9 (Burden)**.** Burden is the total support demand required to realise and maintain a candidate transition or regime across its persistence horizon:

$$\mathsf{B}(X \to Y; \ell) \in [0, \infty).$$

Definition 10 (Persistence horizon)**.** The persistence horizon

$$\ell(X \to Y)$$

is the depth, span, interval, or regime-specific extent over which coherence must be maintained for the branch to count as genuine.

Definition 11 (Reserve margin)**.** The reserve margin is

$$\Pi(X \to Y) := \mathsf{S}(X \to Y; \ell) - \mathsf{B}(X \to Y; \ell).$$

Positive reserve indicates support beyond minimum burden, zero reserve indicates marginal support, and negative reserve indicates inadmissibility.

7 Admissibility as an indicator law

We now formalise admissibility in a non-duplicative way.

Definition 12 (Admissibility indicator)**.** For a candidate transition $X \to Y$, define

$$\mathcal{A}(X \to Y) := \mathbf{1}\Big[\Pi(X \to Y) \geq \pi_{\min} \ \wedge \ \mathsf{Cl}(Y) \geq \mathsf{Cl}_{\min}\Big],$$

where $\pi_{\min} \geq 0$ is the minimum reserve requirement and $\mathsf{Cl}_{\min} \geq 0$ is the minimum closure threshold.

Remark 2. Because

$$\Pi(X \to Y) = \mathsf{S}(X \to Y; \ell) - \mathsf{B}(X \to Y; \ell),$$

the condition $\Pi(X \to Y) \geq \pi_{\min}$ with $\pi_{\min} \geq 0$ already subsumes raw burden feasibility. There is therefore no need to state $\mathsf{B} \leq \mathsf{S}$ separately inside the indicator law.

Definition 13 (Admissible transition)**.** Write

$$X \rightsquigarrow_A Y$$

if and only if $\mathcal{A}(X \to Y) = 1$.

Principle 5 (Admissibility principle)**.** A formally definable transition is structurally real only when it is supportable across its required persistence horizon while remaining above the required closure threshold.

Proposition 1. *If $X \rightsquigarrow_A Y$, then necessarily*

$$\Pi(X \to Y) \geq \pi_{\min} \qquad \textit{and} \qquad \mathsf{Cl}(Y) \geq \mathsf{Cl}_{\min}.$$

Proof. Immediate from the definition of $\mathcal{A}(X \to Y)$.

8 Temporal precedence

Time is now defined as order induced by admissible structural commitment.

Definition 14 (Weak temporal precedence)**.** For $X, Y \in \mathcal{X}$, define

$$X \preccurlyeq_T Y$$

if either $X = Y$ or there exists a finite admissible chain

$$X = X_0 \rightsquigarrow_A X_1 \rightsquigarrow_A \cdots \rightsquigarrow_A X_n = Y.$$

Definition 15 (Strict temporal precedence)**.** Define

$$X \prec_T Y$$

if there exists a nontrivial finite admissible chain from X to Y of length at least one.

Lemma 3 (Reflexivity)**.** *For every $X \in \mathcal{X}$,*

$$X \preccurlyeq_T X.$$

Proof. Immediate from the definition of weak temporal precedence.

Lemma 4 (Transitivity)**.** *If $X \preccurlyeq_T Y$ and $Y \preccurlyeq_T Z$, then*

$$X \preccurlyeq_T Z.$$

If $X \prec_T Y$ and $Y \prec_T Z$, then

$$X \prec_T Z.$$

Proof. Concatenate the admissible chains witnessing the two relations.

Definition 16 (Admissible equivalence)**.** Define

$$X \sim_A Y \quad \Longleftrightarrow \quad X \preccurlyeq_T Y \ \wedge \ Y \preccurlyeq_T X.$$

Proposition 2. *The relation $\sim_A$ is an equivalence relation on the admissible sector of $\mathcal{X}$.*

Proof. Reflexivity and symmetry are immediate. Transitivity follows from the transitivity of $\preccurlyeq_T$.

Theorem 1 (Preorder theorem)**.** *The relation $\preccurlyeq_T$ is a preorder on the admissible sector of $\mathcal{X}$.*

Proof. By the reflexivity and transitivity lemmas.

Definition 17 (Directional acyclicity)**.** An admissible sector is directionally acyclic if there do not exist distinct states

$$X_0, X_1, \ldots, X_n = X_0$$

forming a nontrivial admissible cycle with strictly positive total closure increment or strictly positive total memory increment around the cycle.

Theorem 2 (Partial order on the directional quotient)**.** *If the admissible sector is directionally acyclic, then the quotient relation induced by $\preccurlyeq_T$ on $\mathcal{X}/\sim_A$ is a partial order.*

Proof. Reflexivity and transitivity are inherited from the preorder. Antisymmetry follows because mutual precedence of distinct equivalence classes would generate a nontrivial direction-bearing cycle, contradicting directional acyclicity.

Principle 6 (Time as admissible refinement order)**.** Temporal order is not bare refinement order. It is refinement order filtered by closure and admissibility.

9 Structural memory

If temporal precedence is to carry direction, a monotone quantity must orient admissible chains. That role is played here by structural memory.

Definition 18 (Residue functional)**.** Let

$$\mathsf{Res} : \mathcal{X} \to [0, \infty)$$

measure persistent internal residue encoded by previous closure-bearing transitions.

Definition 19 (Branch-loss functional)**.** Let

$$\mathsf{Br} : \mathcal{X} \to [0, \infty)$$

measure the degree of irreversible branch-loss, excluded possibility, or path commitment accrued in reaching state X.

Definition 20 (Structural memory)**.** Define structural memory by

$$\mathsf{M}(X) := \alpha_m \mathsf{Cl}(X) + \beta_m \mathsf{Res}(X) + \gamma_m \mathsf{Br}(X),$$

where $\alpha_m, \beta_m, \gamma_m > 0$.

Definition 21 (Memory increment)**.** For a transition $X \to Y$, define

$$\Delta\mathsf{M}(X, Y) := \mathsf{M}(Y) - \mathsf{M}(X).$$

Axiom 1 (Residue monotonicity on irreversible residue-bearing transitions)**.** *If $X \prec_T Y$ through a chain that deposits persistent residue, then*

$$\mathsf{Res}(Y) \geq \mathsf{Res}(X).$$

Axiom 2 (Branch-loss monotonicity on irreversible exclusion events)**.** *If $X \prec_T Y$ through a chain that irreversibly excludes prior admissible branch options, then*

$$\mathsf{Br}(Y) \geq \mathsf{Br}(X).$$

Proposition 3. *Suppose $X \prec_T Y$ and at least one of the following holds:*

i. $\Delta\mathsf{Cl}(X, Y) > 0$,

ii. $\mathsf{Res}(Y) > \mathsf{Res}(X)$,

iii. $\mathsf{Br}(Y) > \mathsf{Br}(X)$.

Then

$$\Delta\mathsf{M}(X, Y) > 0.$$

Proof. Each coefficient $\alpha_m, \beta_m, \gamma_m$ is positive, and the relevant increments are nonnegative with at least one strict increase.

Corollary 1 (Memory monotonicity)**.** *If $X \prec_T Y$ through a chain containing positive irreversible closure commitment, residue deposition, or branch-loss, then*

$$\Delta\mathsf{M}(X, Y) \geq 0,$$

with strict inequality whenever new persistent commitment occurs.

10 Irreversibility and directional asymmetry

The arrow of time must be quantified as asymmetry in admissible persistence rather than merely asserted.

Definition 22 (Support surplus)**.** For a candidate transition $X \to Y$, define

$$\Sigma(X, Y; \ell) := \Pi(X \to Y).$$

Definition 23 (Directional asymmetry)**.** For candidate states X, Y, define

$$\Omega(X, Y; \ell) := \Sigma(X, Y; \ell) - \Sigma(Y, X; \ell).$$

Thus:

$$\begin{aligned} \Omega(X, Y; \ell) &> 0 \quad \text{forward-favoured}, \\ \Omega(X, Y; \ell) &= 0 \quad \text{surplus-symmetric}, \\ \Omega(X, Y; \ell) &< 0 \quad \text{reverse-favoured}. \end{aligned}$$

Principle 7 (Directional asymmetry principle)**.** A transition is temporally directional when admissibility is not invariant under reversal and that non-invariance is reflected by positive directional asymmetry together with nondecreasing structural memory.

Proposition 4 (Asymmetry-to-inadmissibility bridge)**.** *Let $X, Y \in \mathcal{X}$. Assume:*

i. $X \rightsquigarrow_A Y$,

ii. $\Pi(X \to Y) \geq \pi_{\min}$,

iii. $\Omega(X, Y; \ell) > \Pi(X \to Y) - \pi_{\min}$.

Then

$$\Pi(Y \to X) < \pi_{\min},$$

and hence

$$\mathcal{A}(Y \to X) = 0.$$

Proof. By definition,

$$\Omega(X, Y; \ell) = \Pi(X \to Y) - \Pi(Y \to X).$$

If

$$\Omega(X, Y; \ell) > \Pi(X \to Y) - \pi_{\min},$$

then rearranging gives

$$\Pi(Y \to X) < \pi_{\min}.$$

Therefore the reverse transition fails the admissibility condition.

Theorem 3 (Irreversibility theorem)**.** *Let $X, Y \in \mathcal{X}$. Suppose:*

i. $X \rightsquigarrow_A Y$,

ii. $\Delta\mathsf{Cl}(X, Y) > 0$ *or* $\Delta\mathsf{M}(X, Y) > 0$,

iii. either the asymmetry-to-inadmissibility bridge applies or every reverse chain from Y to X violates the reserve or closure condition.

Then

$$X \prec_T Y \qquad \textit{and} \qquad \neg(Y \prec_T X),$$

so the transition contributes nontrivially to the temporal arrow.

Proof. Since $X \rightsquigarrow_A Y$, there exists a nontrivial admissible chain from X to Y, hence $X \prec_T Y$. By assumption, the reverse direction fails either directly by reserve asymmetry or more generally because every reverse chain violates admissibility. Therefore $\neg(Y \prec_T X)$.

11 Duration as intrinsic path measure

Time becomes measurable once admissible direction-bearing paths are assigned intrinsic weights.

Definition 24 (Admissible path)**.** An admissible path from X to Y is a finite chain

$$\gamma : \quad X = X_0 \rightsquigarrow_A X_1 \rightsquigarrow_A \cdots \rightsquigarrow_A X_n = Y.$$

Definition 25 (Transition difficulty)**.** Let

$$\Xi : \mathcal{X} \times \mathcal{X} \to [0, \infty)$$

measure structural reconfiguration difficulty not already absorbed into closure or memory changes.

Definition 26 (Local transition weight)**.** For a nontrivial admissible transition $X \rightsquigarrow_A Y$, define

$$w(X,Y) := a\,\Delta\mathsf{Cl}(X,Y) + b\,\Delta\mathsf{M}(X,Y) + c\,\frac{\mathsf{B}(X \to Y;\ell)}{\mathsf{S}(X \to Y;\ell)} + d\,\Xi(X,Y),$$

where $a, b, c, d > 0$.

Remark 3. The burden ratio is well-defined on admissible transitions because support must be positive whenever reserve is finite and the transition is supportable.

Lemma 5. *If $X \rightsquigarrow_A Y$ is nontrivial and at least one of*

$$\Delta\mathsf{Cl}(X,Y),\ \Delta\mathsf{M}(X,Y),\ \Xi(X,Y)$$

is positive, then

$$w(X,Y) > 0.$$

Proof. All coefficients are positive, and each term is nonnegative on admissible transitions.

Definition 27 (Intrinsic duration of a path)**.** For an admissible path

$$\gamma:\ X_0 \rightsquigarrow_A X_1 \rightsquigarrow_A \cdots \rightsquigarrow_A X_n,$$

define

$$\tau(\gamma) := \sum_{i=0}^{n-1} w(X_i, X_{i+1}).$$

Proposition 5 (Concatenation additivity)**.** *If γ_1 is an admissible path from X to Y and γ_2 is an admissible path from Y to Z, then*

$$\tau(\gamma_1 * \gamma_2) = \tau(\gamma_1) + \tau(\gamma_2),$$

where $\gamma_1 * \gamma_2$ denotes path concatenation.

Proof. Immediate from the definition of τ as a sum over successive path segments.

Proposition 6. *If a path contains no nontrivial admissible transition, then its intrinsic duration is zero.*

Proof. There are no positive-weight segments to sum.

Remark 4. Distinct admissible paths between the same endpoints need not have equal intrinsic duration. This is a feature, not a flaw. Duration is path-sensitive because structural commitment is path-sensitive.

12 Continuous analogue

The discrete formulation is primary for this paper, but its continuous analogue is useful.

Let $\gamma : [0,1] \to \mathcal{X}$ be a regular admissible trajectory through a suitably topologised or manifold-like sector of state space. If ω is a positive transition-density functional along the path, define

$$\tau(\gamma) = \int_0^1 \omega(\gamma(s), \dot\gamma(s))\, ds.$$

One may view the discrete weight $w(X_i, X_{i+1})$ as a Riemann-sum approximation to this integral. The continuous representation is not required for the core argument but shows that the path-measure interpretation is not tied to a purely discrete exposition.

13 Clocks and coarse-grained recurrence

A clock must recur locally without violating the exclusion of nontrivial direction-bearing cycles. This is achieved by distinguishing literal identity from coarse-grained observational equivalence.

Definition 28 (Clock observational equivalence)**.** For a clock subsystem K, define an equivalence relation

$$X \sim_K Y$$

when X and Y are observationally indistinguishable at the scale of the clock readout, even if they differ microscopically in accumulated residue or ambient embedding.

Definition 29 (Clock subsystem)**.** A clock subsystem is a closed or near-closed subsystem admitting a sequence

$$X_0^K \rightsquigarrow_A X_1^K \rightsquigarrow_A \cdots \rightsquigarrow_A X_m^K$$

such that

$$X_m^K \sim_K X_0^K$$

and the corresponding local transition weights are approximately uniform:

$$w_K(X_i^K, X_{i+1}^K) \approx \delta\tau_K \qquad \text{for all } i.$$

Remark 5. This is coarse-grained recurrence, not literal global cyclicity. There is therefore no conflict with the exclusion of nontrivial direction-bearing cycles in the full state description.

Definition 30 (Clock reading)**.** If n coarse-grained recurrent cycles have occurred in subsystem K, define

$$T_K(n) := n\,\delta\tau_K.$$

Theorem 4 (Clock emergence theorem)**.** *Any sufficiently stable subsystem with coarse-grained recurrent admissible behaviour and approximately uniform local transition weights realises an emergent clock.*

Proof. Each recurrent cycle contributes approximately the same intrinsic duration increment $\delta\tau_K$. Counting successive cycles therefore yields a stable reading variable.

Principle 8 (Clock principle)**.** Clocks do not detect a universal temporal substance. They count approximately uniform local admissible reconfiguration cycles.

14 Time dilation from closure density

We now connect the present account to relativistic phenomenology without granting time primitive ontological status.

Let ρ_C denote local closure density or local closure-commitment environment. Let

$$\delta\tau_K(\rho_C)$$

denote the intrinsic duration cost per coarse-grained clock cycle in subsystem K.

Definition 31 (Closure-density rate law)**.** Assume

$$\delta\tau_K(\rho_C) = \delta\tau_0\, F(\rho_C),$$

where $\delta\tau_0 > 0$ is a reference cycle increment and

$$F(\rho_C) > 0, \qquad F'(\rho_C) > 0.$$

The local cycle frequency is therefore

$$\nu_K(\rho_C) = \frac{1}{\delta\tau_K(\rho_C)} = \frac{1}{\delta\tau_0\, F(\rho_C)}.$$

Proposition 7 (Closure-density slowdown)**.** *If*

$$\rho_{C,1} > \rho_{C,2},$$

then

$$\delta\tau_K(\rho_{C,1}) > \delta\tau_K(\rho_{C,2}) \qquad \textit{and} \qquad \nu_K(\rho_{C,1}) < \nu_K(\rho_{C,2}).$$

Proof. Since $F'(\rho_C) > 0$, larger closure density implies larger $F(\rho_C)$ and hence larger $\delta\tau_K(\rho_C)$. Taking reciprocals yields the frequency relation.

Corollary 2. *For two clock subsystems in different closure-density environments,*

$$\frac{\nu_K(\rho_{C,1})}{\nu_K(\rho_{C,2})} = \frac{F(\rho_{C,2})}{F(\rho_{C,1})}.$$

Principle 9 (Closure-density dilation)**.** Clock slowdown in high-closure regimes is not the deformation of a primitive temporal fluid. It is the increase of intrinsic local transition cost per stable clock cycle.

15 Structural age

Because intrinsic duration is path-sensitive, structural age must also be path-aware.

Let X_0 denote the earliest meaningful closure-bearing reference state for the regime under study.

Definition 32 (Path family)**.** Let

$$\mathcal{P}(X_0, X)$$

denote the family of admissible paths from X_0 to X.

Definition 33 (Minimal structural age)**.** Define

$$\mathfrak{A}_{\min}(X) := \inf_{\gamma \in \mathcal{P}(X_0,X)} \tau(\gamma).$$

Definition 34 (Canonical structural age)**.** If a canonical developmental path $\gamma^\star_{X_0 \to X}$ is specified by additional physical or ontological criteria, define

$$\mathfrak{A}_{\mathrm{can}}(X) := \tau(\gamma^\star_{X_0 \to X}).$$

Definition 35 (Expected structural age)**.** If μ_X is a probability measure on $\mathcal{P}(X_0, X)$, define

$$\mathfrak{A}_{\exp}(X) := \int_{\mathcal{P}(X_0,X)} \tau(\gamma)\, d\mu_X(\gamma).$$

Remark 6. A single undeclared structural-age functional would conceal path dependence. The present three-form construction makes that dependence explicit and mathematically controlled.

16 Pre-temporal regime

The notion of "before time" can now be stated cleanly.

Definition 36 (Pre-temporal regime)**.** A regime is pre-temporal if it lies below the minimum structural conditions required for meaningful temporal precedence, namely below the thresholds needed for closure-bearing admissible ordering.

Principle 10 (Pre-temporal malformation)**.** Below the threshold at which admissible closure-bearing order becomes well-defined, temporal predicates are not false but malformed.

Thus the earliest meaningful beginning in Darke Theory is not an earliest instant in a pre-existing temporal sequence. It is the first regime in which persistent structural order becomes well-posed.

17 Synthesis results

We now collect the formal content into a compact synthesis.

Definition 37 (Temporality-bearing Darke-Theoretic system)**.** A temporality-bearing system is the data

$$(\mathcal{X}, \preceq, \mathsf{D}, \mathsf{Cl}, \mathcal{A}, \mathsf{S}, \mathsf{B}, \ell, \Pi, \mathsf{M}).$$

Theorem 5 (Derived temporal order theorem)**.** *Suppose:*

i. $(\mathcal{X}, \preceq, \mathsf{D})$ *is a recursive refinement system,*

ii. a nonempty closed sector $\mathcal{C}_{\text{crit}}$ *exists,*

iii. admissibility is given by the indicator law above,

iv. temporal precedence is defined by admissible reachability.

Then $\preccurlyeq_T$ *is a preorder on the admissible sector of* $\mathcal{X}$*, and hence furnishes a derived temporal order.*

Proof. By the reflexivity and transitivity lemmas, $\preccurlyeq_T$ is a preorder. Since it is generated by admissible refinement reachability among closure-bearing states, it furnishes the intended derived temporal order.

Proposition 8 (Arrow synthesis)**.** *Suppose memory is monotone on irreversible admissible chains and reverse admissibility fails whenever the asymmetry-to-inadmissibility bridge applies. Then the temporal order generated by* $\preccurlyeq_T$ *carries an intrinsic arrow on the direction-bearing sector.*

Proof. Monotone memory supplies orientation, while reverse inadmissibility excludes symmetry of traversal for direction-bearing transitions.

Proposition 9 (Duration synthesis)**.** *If positive transition weights are assigned to nontrivial admissible steps, then intrinsic duration exists as an additive path measure on admissible chains.*

Proof. The definition of $\tau(\gamma)$ provides existence. Additivity follows from path concatenation.

Proposition 10 (Clock synthesis)**.** *Any stable subsystem exhibiting coarse-grained recurrent admissible behaviour with approximately uniform cycle weights realises an emergent clock.*

Proof. This is exactly the content of the clock emergence theorem.

Proposition 11 (Closure-density dilation synthesis)**.** *If the per-cycle duration law satisfies*

$$\delta\tau_K(\rho_C) = \delta\tau_0\, F(\rho_C) \quad \textit{with} \quad F'(\rho_C) > 0,$$

then clocks in higher-closure-density regimes run more slowly than clocks in lower-closure-density regimes.

Proof. Immediate from the closure-density slowdown proposition.

18 Interpretive consequences

The strengthened mathematics clarifies the paper's interpretive content.

18.1 Time is derived, not assumed

Time is reconstructed from admissible, closure-bearing refinement rather than presupposed as an external axis.

18.2 Irreversibility is structural, not merely statistical

The arrow arises from reserve asymmetry and structural memory, not merely from a generic appeal to disorder.

18.3 Memory becomes mathematically load-bearing

Because memory is decomposed into closure, residue, and branch-loss terms, temporal direction is anchored in explicit structural commitment.

18.4 Clock behaviour remains empirical but is re-ontologised

Observed clock regularity and slowdown are preserved, but their meaning is shifted from primitive time deformation to variation in local admissible reconfiguration cost.

18.5 Cosmic age becomes structural age

Late-time cosmology should not privilege a single global elapsed parameter without recognising closure stratification and path dependence.

19 Scope

This paper is a foundational reconstruction of temporality, not a complete dynamical theory of every physical regime. It does not attempt to derive all effective equations of motion from first principles in one step. Its claim is narrower and deeper: that the ontological status of time is misplaced if treated as primitive, and that Darke Theory contains the machinery required to rebuild it from recursive differentiation, closure, admissibility, memory, and burdened persistence.

20 Conclusion

The central demand of a foundational theory is discipline. If time is to be explained, it may not be smuggled in as primitive. Darke Theory respects that demand by beginning not from temporal flow but from recursive differentiation. Persistent structure emerges only where recursive refinement achieves closure and survives admissibility filtering. Once that framework is taken seriously, temporality must be rebuilt rather than assumed.

The present paper has carried out that reconstruction in strengthened mathematical form.

Closure has been formalised as a graded functional with constrained subcomponents. Admissibility has been defined as an indicator law governed by reserve and closure threshold. Temporal precedence has been defined as admissible refinement reachability and given explicit order-theoretic structure. Structural memory has been decomposed into closure, residue, and branch-loss terms, with monotonic growth inherited through irreversibility-bearing transitions. Reverse inadmissibility has been tied more tightly to directional asymmetry. Duration has been defined as additive intrinsic path measure with an explicit transition-weight ansatz. Clocks have been reconstructed as coarse-grained recurrent counters of approximately uniform local admissible cycles. Time dilation has been reinterpreted as closure-density dependence of local transition cost. Structural age has been made path-aware.

The compressed result is:

> **Time in Darke Theory is the measurable order of admissible structural commitment, directed by irreversible memory growth and quantified by intrinsic path measure.**

Time is therefore not the background in which persistence occurs. It is what persistence looks like once closure, admissibility, and memory become mathematically real.

A Appendix: constraint summary

For clarity, the principal structural constraints used in the paper are collected here.

1. Closure subfunctionals satisfy

$$I_{\text{self}},\ I_{\text{rec}},\ I_{\text{supp}},\ I_{\text{pert}} \in [0,1].$$

2. Closure is bounded above by the coefficient sum:

$$0 \leq \mathsf{Cl}(X) \leq \alpha_c + \beta_c + \gamma_c + \delta_c.$$

3. Admissibility is determined by

$$\Pi(X \to Y) \geq \pi_{\min} \quad \text{and} \quad \mathsf{Cl}(Y) \geq \mathsf{Cl}_{\min}.$$

4. Structural memory is given by

$$\mathsf{M}(X) = \alpha_m \mathsf{Cl}(X) + \beta_m \mathsf{Res}(X) + \gamma_m \mathsf{Br}(X),$$

   with positive coefficients.

5. Path duration is additive:

$$\tau(\gamma_1 * \gamma_2) = \tau(\gamma_1) + \tau(\gamma_2).$$

6. Clock slowdown is controlled by

$$\delta\tau_K(\rho_C) = \delta\tau_0\, F(\rho_C), \qquad F'(\rho_C) > 0.$$

B Appendix: expected monotonicity conditions

The main text treats residue and branch-loss monotonicity as structural assumptions on the relevant irreversible transition classes. It is useful to record sufficient conditions under which those assumptions are naturally expected.

Proposition 12 (Expected residue monotonicity)**.** *Let $X \prec_T Y$ through an admissible chain*

$$X = X_0 \rightsquigarrow_A X_1 \rightsquigarrow_A \cdots \rightsquigarrow_A X_n = Y.$$

Assume that along each step:

i. persistent closure-bearing reconfiguration deposits non-negative residual structure,

ii. no admissible operation on the chain removes previously deposited residue at zero net closure or support cost,

iii. residue contributions are additive or at least monotone under chain concatenation.

Then

$$\mathsf{Res}(Y) \geq \mathsf{Res}(X).$$

If at least one step deposits strictly positive new residue, then

$$\mathsf{Res}(Y) > \mathsf{Res}(X).$$

Proof. By assumptions (i) and (iii), each step contributes a non-negative increment to the total residue, while assumption (ii) excludes lossless erasure along the admissible chain. Hence the total residue cannot decrease from X to Y. Strict increase follows if at least one step contributes a strictly positive residue increment.

Proposition 13 (Expected branch-loss monotonicity)**.** *Let $X \prec_T Y$ through an admissible chain*

$$X = X_0 \rightsquigarrow_A X_1 \rightsquigarrow_A \cdots \rightsquigarrow_A X_n = Y.$$

Assume that along each step:

i. closure commitment or support allocation may eliminate previously admissible alternative continuation branches,

ii. once such excluded branches fall below admissibility threshold they are not restored without traversing a reverse-inadmissible reconstruction,

iii. branch-loss accumulates monotonically under admissible composition.

Then

$$\mathsf{Br}(Y) \geq \mathsf{Br}(X).$$

If at least one step irreversibly excludes a previously admissible branch, then

$$\mathsf{Br}(Y) > \mathsf{Br}(X).$$

Proof. Assumption (i) allows admissible transitions to reduce the set of viable future branches. Assumption (ii) prevents those lost branches from being recovered within the same admissible forward sector. Assumption (iii) ensures cumulative non-decrease under composition. Hence branch-loss is monotone along the chain, with strict increase whenever a genuinely admissible alternative is irreversibly excluded.

Corollary 3 (Expected memory monotonicity)**.** *Under the sufficient conditions of the two propositions above, and with*

$$\mathsf{M}(X) = \alpha_m \mathsf{Cl}(X) + \beta_m \mathsf{Res}(X) + \gamma_m \mathsf{Br}(X), \qquad \alpha_m, \beta_m, \gamma_m > 0,$$

any admissible chain with non-negative closure increment and non-decreasing residue and branch-loss satisfies

$$\mathsf{M}(Y) \geq \mathsf{M}(X).$$

If at least one component increases strictly, then

$$\mathsf{M}(Y) > \mathsf{M}(X).$$

Proof. Immediate from positivity of the coefficients and monotonicity of the component functionals.

References

[1] James Alan Darke, *Classification of Stable Directed Refinement Systems*, 2026.

[2] James Alan Darke, *Recursive Differentiation and Structural Closure*, 2026.

[3] James Alan Darke, *Darke Theory Ontological Foundation*, 2026.

[4] James Alan Darke, *Darke Theory Core*, 2026.

[5] James Alan Darke, *Infinite Codimension Under Finite Symmetry*, 2026.

[6] James Alan Darke, *Recursive Differentiation and the Structural Origin of Quantum Mechanics*, 2026.

[7] James Alan Darke, *The Limits of Darke Theory*, 2026.

[8] James Alan Darke, *The Critical Closure Stability Theorem*, 2026.

[9] James Alan Darke, *Darke Theory: From Recursive Differentiation to Admissibility, Quantum Structure and Cross-Domain Exclusion*, 2026.

[10] James Alan Darke, *The Coherence Burden Principle*, 2026.

[11] James Alan Darke, *Relativistic Closure and the Curvature Bridge*, 2026.

[12] James Alan Darke, *Directional and Void Conditioned Variations in the Local Hubble Relation*, 2026.